Diseases of Potato

1. Late blight - *Phytophthora infestans*
2. Early blight - *Alternaria solani*
3. Sclerotium rot - *Sclerotium rolfsii*
4. Wart - *Synchytrium endobioticum*
5. Dry rot - *Fusarium coerulem*
6. Black Scurf and canker - *Rhizoctonia solani*

Diseases of Potato

Bacterial diseases

1. Common scab - *Streptomyces scabies*
2. Black leg and soft rot - *Erwinia carotovora*, *E. c.* sub sp. *carotovora*, *E. chrysanthemi*
3. Brown rot and bacterial wilt - *Ralstonia solanacearum*

Virus and Viroid diseases - 30 viruses - 5 or 6 – India

1. Mosaic - *Potato virus X* (PV X), *Potato virus Y* (PV Y) and *Potato virus S* (PV S)
2. Leaf roll/ leaf curl phloem necrosis - *Potato leaf roll virus* (PLRV)
3. Spindle tuber - *Potato spindle tuber viroid* (PSTVd)

Phytoplasma disease

1. Witches' broom - Phytoplasma

Late blight of potato- *Phytophthora infestans*

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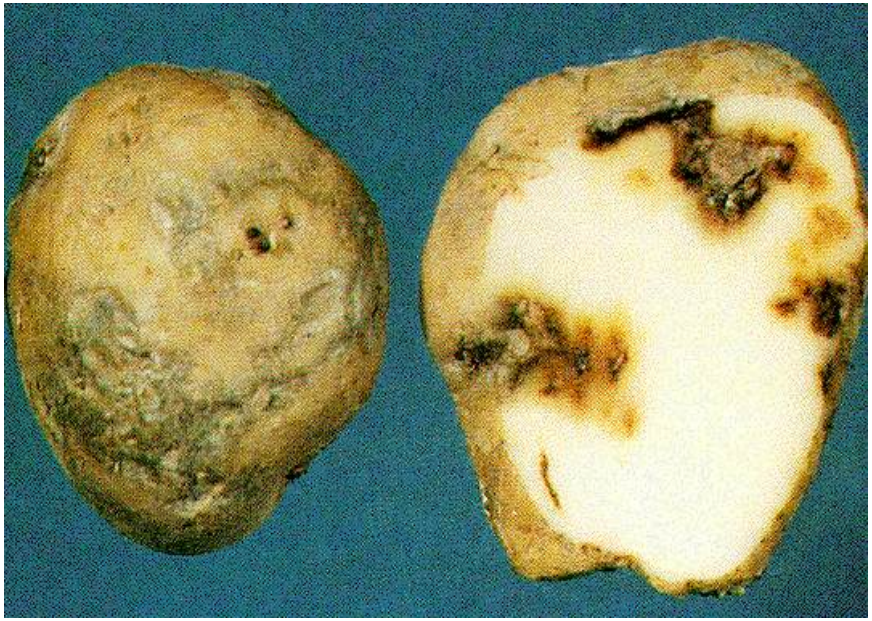
Late blight of potato- *Phytophthora infestans*



Late blight of potato- *Phytophthora infestans*

Irish famine 1845-46

First reported in1870 from Nilgiri hills, Darjeeling, Shimla

Symptoms

- Initially starts from **leaf tips or margins to inward**
- **Small faded green patches** - brown to purplish black spots
- Downy growth on lower surface - morning hours
- On stems, petioles - light to dark brown lesions, necrotic
- Tender growing tips - withering and death
- **Progressive defoliation and collapse of plants** even in 2 weeks the crop may be damaged
- **Purplish brown spots** on skin of **tubers**
- On cutting, the **affected tubers show rusty brown necrosis** spreading from surface to the centre
- Decay of **plant parts emits foul smell**

Late blight of potato- *Phytophthora infestans*

Disease development:

Dormant mycelium in the infected tubers in storage

Cull plies and volunteer plants near cold stores

Oospores formation has been reported from hilly areas- HP

Blight forecasting :

Blitecast - North eastern USA for fungicide spraying

Prediction of outbreaks: in hills

Seven day moving rainfall of 30 mm

Seven day mean temp of 23.9^{o}C or less } 3weeks

Hourly temp. ranging between 10-20°C with 80% relative humidity for 18 hrs for 2 consecutive days - **the disease appear with in a week.**

Late blight of potato- *Phytophthora infestans*

Management: Sanitation

- Regulatory measures
- Select healthy tubers for planting
- Delayed harvesting
- **High ridging to about 10-15cm height reduces tuber infection**
- Grow resistant varieties **Kufri Jyothi, Kufri Badshah, Kufri Jeevan, Kufri Sherpa, etc.**
- **Resistant sources: *Solanum demissum and S. phureja***

Prophylactic measures

- Metalaxyl (0.1%) or mancozeb (0.25%) or chlorothalonil (0.2%) or BM (1%) at 7 to 10 days intervals in the hills and 10 to15 days intervals in plains.
- Dip sprouted tubers in 0.2% metalaxyl for 30 min

Early blight *Alternaria solani*

Early blight *Alternaria solani*

Symptoms

- Primarily **attacks foliage, uncommon on stems, tubers**
- Older leaves infected **first-dark brown oval to angular spots**
- Enlarge with **concentric rings**
- In dry weather the necrotic part of the lesion fall off, on petioles **dark brown elongated lesions**
- **On tubers dark sunken lesions surrounded by raised border with dry and leathery flesh underneath.**

Early blight *Alternaria solani*

Infected plant debris, weed hosts and tomato-Primary source

Management:

- Destruction of infected plant debris
- Harvesting of the tubers after maturity
- Proper drying before storage
- Fungicide spraying
- **Res. Var. Kufri sindhuri**
- **Rest. Source*: S. phureja, S. chacoense***

Potato Wart – *Synchytrium endobioticum*

Potato Wart – *Synchytrium endobioticum*

Introduced along with variety, Furore from Demark in Darjeeling dt, WB. It is restricted to Darjeeling and adjoining areas of Nepal in India

Symptoms

- **White tumour** like outgrowths (warts) on stems, stolons and tubers
- Host cells at the point of infection are **hypertrophied**
- **Epidermal cells of tubers** are infected by the fungus
- **White granular swellings** form on the eyes of tubers - normally soft, pulpy, spherical initially (look like tuber) later changes to **green**
- As the crop matures, warts become **black and rot** due secondary infection
- **Infected auxillary bud or the leaf is transformed into cock's comb like greenish yellow outgrowth**

Life cycle

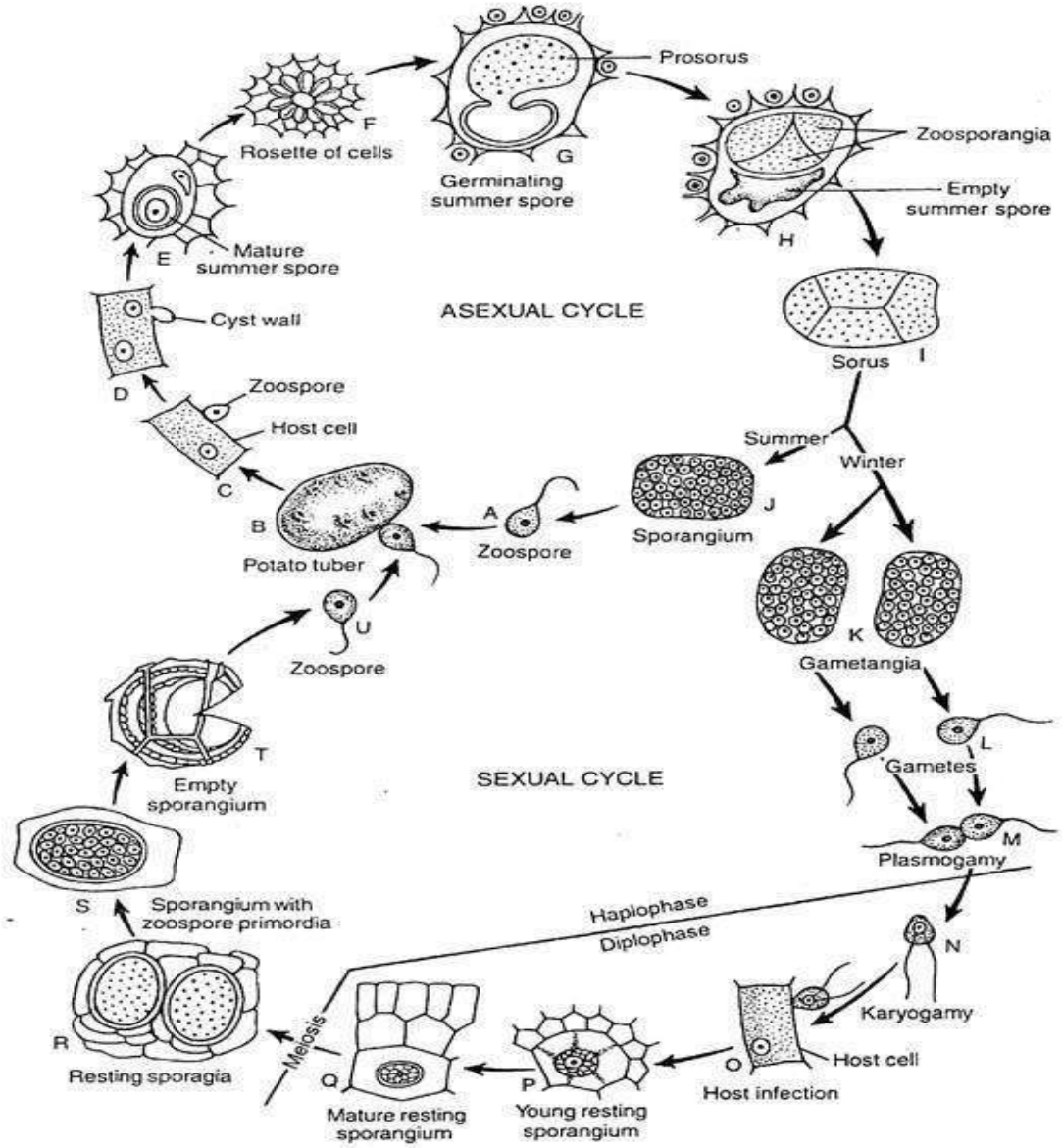


Fig. 4.17 : Life cycle of *Synchytrium endobiotium*

Kingdom: Fungi
Phylum: *Chytridiomycota*
Class: *Chytridiomycetes*
Order: *Chytridiales*
Family: *Synchytriaceae*
Synchytrium endobioticum

Potato Wart – *Synchytrium endobioticum*

- **Thallus is unicellular, endobiotic and holocarpic.**
- Warts on host contain **resting sporangia**.
- **Thallus behaves as a prosorus.**
- **Hypertrophy and hyperplasia takes place, as a result, out growths appear on tubers.**
- The fungus *S. endobioticum* **transmits potato virus -x and potato mop top virus**

Potato Wart – *Synchytrium endobioticum*

Resting spores in soil, plant debris

Management

•**Strict quarantine measures** - movement of potatoes from Darjeeling dist., WB

• **Grow resistant varieties - kufri Jyoti, kufri Jeevan and kufri Muthu**

•**Immune variety: F3977**

•Crop rotation with non host crops for **8-10 years**

•Steam sterilizaion of soil

Potato dry rot –*Fusarium coerulem*

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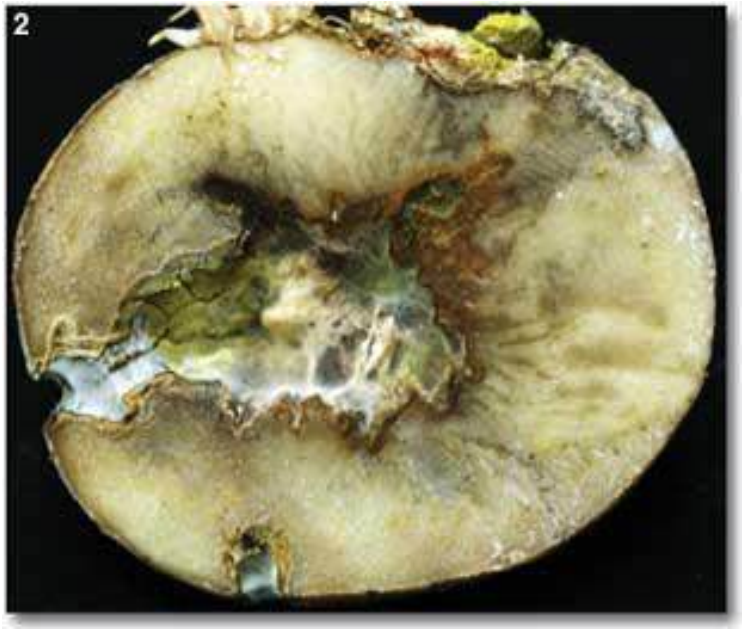


Potato dry rot –*Fusarium coerulem*

More common in country (general) stores

Symptoms

- Dark spots on **skin, sunken, wrinkled with irregular concentric rings**
- Internal tissues becomes **brown, sunken with cavities**
- **Rotting progress into the whole tubers**
- Tubers then becomes **dry, hard, shriveled, light in weight**

Potato dry rot –*Fusarium coerulem*

Seed borne

Management

- Use of disease free seed tubers
- **Tuber treatment**
- **Disinfection of boxes, bags** with copper sulphate 2% /formalin 5% solution

Black Scurf and canker- *Rhizoctonia solani*

Figure 1.

Figure 2.

Figure 3.

Figure 4.

Figure 5.

Figure 6.

Black Scurf and canker- *Rhizoctonia solani*

Sunken lesions on stem

Rotten stem

Black Scurf and canker- *Rhizoctonia solani*

Tuber russet

Black speck on tubers

Black Scurf and canker- *Rhizoctonia solani*

- **Black scurf phase is more common** than stem canker in India.
- Damage is most severe **at cold temperatures**, **emergence of stems and stolons** from the tuber and growth is relatively slow than the pathogen.
- **Wet soils and excessive soil moisture** slows plant development and favors the pathogen
- **Rhizoctonia does not cause seed decay**; its damage is limited to sprouts and stolons.
- **Poor stands and stunted plants** can also be caused by **blackleg,** a bacterial disease that initiates from the seed tuber and progresses up the stems, causing a wet, sometimes slimy, rot.
- Poor stand may be mistaken for **seed tuber decay, caused by Fusarium species or soft rot bacteria**, unless the plants are uprooted and examined.

Black Scurf and canker- *Rhizoctonia solani*

- In contrast, Rhizoctonia lesions are always dry and usually sunken.
- In case of severe disease, **lesions on the shoots girdle the shoot partially or completely, thus sprouts may be stunted / not emerge from the soil.**
- **Stolon cankers reduce tuber numbers and size** and are identical to shoot cankers in appearance.
- Late season occurrence result in cankers on stolons and stems affects starch translocation.
- If **stolons and underground stems are severely infected tubers deformed.**

Black Scurf and canker- *Rhizoctonia solani*

- In such case, small, green tubers, called aerial tubers, may form on the stem above the soil.
- **Formation of aerial tubers** may indicate that under ground tubers of poor quality and not marketable
- At the end of the growing season, the fungus produces its sexual state, ***Thanetephorus cucumeris***, on stems just above the soil line.
- It appears as a **superficial delicate white mat** which is easily removed.
- The fungus does not damage the tissue beneath this mycelium.

Black Scurf and canker- *Rhizoctonia solani*

Disease cycle and epidemiology:

- The fungus perpetuates in the form of sclerotia on the seed tubers or in the soil.
- On germination in the presence of moisture, the sclerotia produce mycelium which may cause infection on the sprouts, stem or young tubers.
- The fungus can survive in soil for many years.
- The fungus although has a wide host range but shows some level of specificity depending on the host crops.
- The strains infecting members of graminicolous crops do not generally cause infections on solanaceous crops including potato and *vice versa*.

Black Scurf and canker- *Rhizoctonia solani*

Disease cycle and epidemiology:

- Soil moisture and temperature are the two important factors affecting scurf development.
- Low temperature and high soil moisture are conducive for infection on sprouts and stems.
- This phase is not common on early autumn crop in North Indian plains as the temperature remains high.
- The more common phase is black scurf where the sclerotia formation is favoured at temperature 28°C.
- The number of sclerotia on tubers varies with soil temperature and moisture.
- Black scurf on tubers is **more common in crops grown in sandy to sandy loam and moderately wet soils.**

Black Scurf and canker- *Rhizoctonia solani*

Management:

- Planting of healthy and disease free tubers is helpful in reducing the incidence of the disease.
- Two to four years rotation with cereals, brassicas and legumes is helpful for the management of this disease.
- The increase in organic matter content of the soil helps in reducing the population of the fungus through enhanced activity of the antagonist microorganisms.
- The normal unsprouted tubers should be dipped in carbendazim (0.1%) for 10 minutes.
- The seed after treatment should be dried under shade by spreading on the floor of the cold store.

Common scab /American scab/Corky scab
Streptomyces scabies

Common scab /American scab/Corky scab
Streptomyces scabies

Plains of Punjab, Bihar, Haryana, UP

Prevalent in **neutral, slightly alkaline soils** under dry conditions

Symptoms:

- **Young tubers** are more susceptible to infection
- Small brownish and **slightly raised spots on tubers**
- Spots **enlarge, coalesce and become corky**
- Lesions typically possess a **raised margin due to abnormal proliferation of epidermal cells** and slightly depressed center

Common scab /American scab/Corky scab
Streptomyces scabies

Symptoms:

Characteristic symptoms

- **Russet scab -** superficial tan to brown corky lesions
- **Pitted scab** - lesions with depressions beneath the tuber
- **Raised scab** appears as cushion like warty lesions

Common scab /American scab/Corky scab
Streptomyces scabies

Primary source – Infected tubers & soil, **FYM** if infected parts are fed to the cattle

Management

- Use of disease free tubers
- Irrigation at 8 days interval from tuberisation until maturity
- Tuber treat. 3% boric acid
- **Crop rotation with wheat-oat or potato-onion-maize (4yrs)**

Common scab /American scab/Corky scab
Streptomyces scabies

Management contd…

- Hold the soil **pH at about 5.3 with sulphur**
- Green manuring before planting potato encourages antagonistic microbes
- Dipping of infected tubers in 3% boric acid for 30 min
- Soil application of formalin 5%
- Balanced fertilizer application

Blackleg and soft rot *P. carotovorum* subsp. *carotovorum* (syn. *E. c*. subsp. *carotovora*)

Blackleg and soft rot *P. carotovorum* subsp. *carotovorum* (syn. *E. c*. subsp. *carotovora*)

Blackleg symptoms

Potato plant - stem collapse due to blackleg

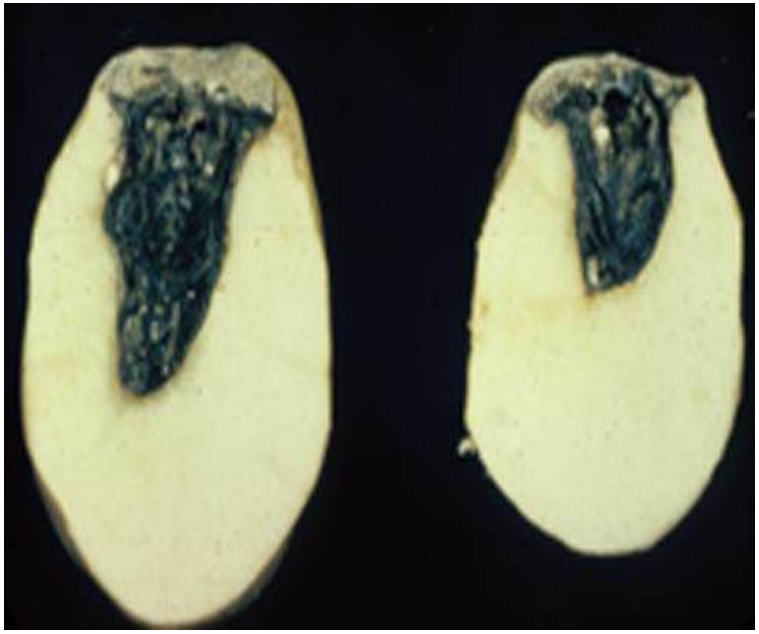
Blackleg and soft rot *P. carotovorum* subsp. *carotovorum* (syn. *E. c*. subsp. *carotovora*)

Black leg symptoms on potato stems

Black lesion at the stem base and aerial tubers on stems.

Blackleg and soft rot *P. carotovorum* subsp. *carotovorum* (syn. *E. c*. subsp. *carotovora*)

Potato plant showing early wilting symptoms of blackleg



Soft rot of potato tuber

Blackleg and soft rot *P. carotovorum* subsp. *carotovorum* (syn. *E. c*. subsp. *carotovora*)

Soft rot

Infected tuber

Blackleg and soft rot *P. carotovorum* subsp. *carotovorum* (syn. *E. c*. subsp. *carotovora*)

Symptoms of blackleg

- Plants are **stunted with a stiff, erect growth habit**.
- The **leaflets tend to roll upward at the margins.**
- Stems of infected plants exhibit an **inky black decay.**
- The base of the **stem is often completely rotted.**
- Plants may wilt.
- In relatively **dry soil, only the pith may show blackening**

Blackleg and soft rot *P. carotovorum* subsp. *carotovorum* (syn. *E. c*. subsp. *carotovora*)

Symptoms of soft rot

- **Soft rot also infects wounded stems and roots**
- Tissues are **wet, cream to tan in color and soft.**
- **Rot begins on the tuber surface and progresses inward.**
- Delineated from healthy tissue by **dark brown or black margins.**
- **Shallow necrotic spots** on the tubers result from infections through lenticels
- **Rotting tissue - odorless** in the early stages of decay, a foul odor due to secondary organisms

Blackleg and soft rot *P. carotovorum* subsp. *carotovorum* (syn. *E. c*. subsp. *carotovora*)

Seed tubers, weeds , plant debris as source of inoculum

Management

- Crop rotation with cereals
- Crop should be planted in well drained areas and at sufficient distances to allow adequate ventilation
- Cleaning and disinfection of warehouses - formaldehyde or copper sulphate
- Humidity and temperatures of warehouses - low

Blackleg and soft rot *P. carotovorum* subsp. *carotovorum* (syn. *E. c.* subsp. *carotovora*)

Management

- Wash tubers with chlorinated water before storage
- Avoid wounding of tubers
- Products to be stored should be dry, dry cut tubers at 12-15°C for 4 days
- Dip cut seed pieces of potato in a solution of Streptocycline 100ppm and $CuSo_4$ 40ppm for 30 min
- Insect control

Bacterial wilt and brown rot - *Ralstonia solanaceraum*

Bacterial wilt and brown rot - *Ralstonia solanaceraum*

Symptoms

- Wilting, stunting and yellowing of foliage collapse plant
- **Browning** of xylem in vascular bundles
- Stems, petioles of the lower leaves and roots become brown and on cutting of infected materials slimy mass of bacteria oozes out

Bacterial wilt and brown rot - *Ralstonia solanaceraum*

Symptoms contd…

- **Brown ring** - in the tubers due to discolouration of vascular bundles
- Formation of **pockets or cavities around vascular bundles in the pith and cortex**
- When pressed slimy bacterial ooze emerge.
- The tubers may rot at harvest or during storage.

Bacterial wilt and brown rot - *Ralstonia solanaceraum*

Survives in plant debris, soil, alternate and collateral hosts

Management

- Crop rotation with maize or soybean for 3 yrs
- Use disease free tubers for sowing
- Deep summer ploughing
- Cut tuber to be kept at 12°C- 4 days
- Treat seed tubers with 0.02% streptocycline for 30 min giving 4mm deep sharp cut
- **Disinfection of cutting knives with sodium hypochlorite (10%)**

Potato leaf roll /leaf curl/phloem necrosis
Potato leaf roll virus (PLRV)

Potato leaf roll /leaf curl/phloem necrosis
Potato leaf roll virus (PLRV)

The lower leaves are severely rolled and leathery to the touch.

The upper leaves become pale, upright, and rolled and show some reddening of the tissue around the leaf edges.

Potato leaf roll /leaf curl/phloem necrosis
Potato leaf roll virus (PLRV)

Potato leaf roll /leaf curl/phloem necrosis
Potato leaf roll virus (PLRV)

Luteovirus- Luteovirus with spherical particles having single stranded RNA

Symptoms

- **Upward rolling of leaves, midrib**
- Later like **tunnel**- stiff, leathery
- Some times with **pink pigmentation**
- Plants stunted, **stiff** with **upright growth**
- More, small tubers/plant
- **Phloem necrosis**, necrosis of tubers in storage

Potato leaf roll virus (PLRV)

Disease cycle and epidemiology

Transmitted through infected tubers and spreads through several species of aphids *Myzus persicae*, *Aphis gossypii*

It is more severe in warm climate.

Potato leaf roll virus (PLRV)

Management

- Disease free seed tubers for planting
- Rouging of diseased plants
- Aphid control- through raising barrier crops and spraying of insecticides

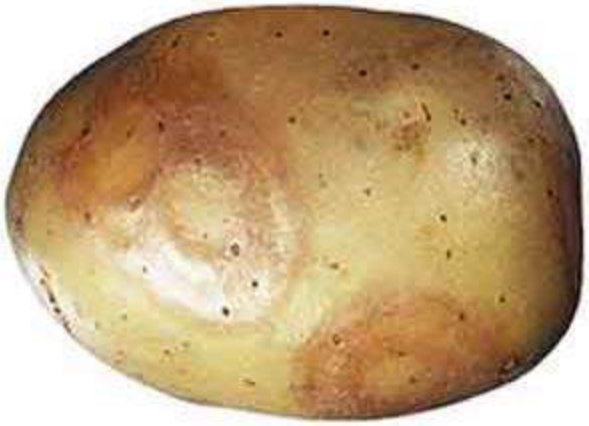
Mosaic disease - *Potato virus X* (PV X), *Potato virus Y* (PV Y) and *Potato virus S* (PV S)

Mild mosaic/Interveinal mosaic –*Potato virus* X (PV X)

Symptoms

- Often referred as latent potato mosaic
- Light **yellow mottling** with **slight crinkling**
- Interveinal necrosis of top foliage
- Stunting of diseases plants

Mild mosaic/Interveinal mosaic –*Potato virus* X (PV X)



Severe mosaic - *Potato virus* Y (PV Y) - leaf drop streak

Symptoms

- **Chlorotic streaks** on leaves become necrotic
- Necrosis of leaf veins and **leaf drop streak**
- Interveinal necrosis and stem/petiole necrosis
- Plant remain stunted in growth
- Twisting of the leaves - with PV X and PV A
- Spread by aphids, *M. persicae ,A. gossypii*

Severe mosaic - *Potato virus* Y (PV Y) - leaf drop streak

Rugose mosaic symptom is characteristic of some strains of Potato Virus Y (PVY)

Other strains of PVY produce a general mosaic or a hypersensitive (severe necrotic) reaction.

Severe mosaic - *Potato virus* Y (PV Y)

Leaf drop streak

Severe mosaic - *Potato virus* Y (PV Y)

Tubers with necrotic ring spot - PVY

Potato virus S (PV S)

Symptoms caused by PVS are barely perceptible as mottle or faint vein bending.

Mosaic disease - *Potato virus X* (PV X), *Potato virus Y* (PV Y) and *Potato virus S* (PV S)

Disease cycle and epidemiology:

- Mosaic viruses are transmitted by infected tubers, cutting knife and aphid vectors.

- Many plants of Solanaceae, Chenopodiaceae and Leguminoseae act as alternate and collateral host hosts of this disease.

Mosaic disease - *Potato virus X* (PV X), *Potato virus Y* (PV Y) and *Potato virus S* (PV S)

Management:

- Always use virus free certified seed from reliable sources.
- Rouge out infected plants as soon as these are first noticed in early crop period.
- Keep the aphid population under check by spraying insecticides like Rogor or Metasystox @ 250 ml/acre or apply 5 kg of Thimet 10G/acre at first earthing up.
- Cut the haulms in 4th week of December when aphid infestation starts (20 aphid /100 leaves) and leave the tubers in soil to mature till the end of February.

Potato spindle tuber viroid (PSTVd)

PSTV foliar symptoms show little contrast, making this one of the more difficult diseases to diagnose. The leaflets may be **smaller and curve inwardly, giving a stiff upward growth habit.**

Potato spindle tuber viroid (PSTVd)

Infected tubers may be **spindle or oblong or tend** to be more rounded instead of the normal shape for a given variety.

Potato spindle tuber viroid (PSTVd)

Very small single stranded circular RNA
Reported from Shimla hills in India

Symptoms

- Plants **dark green, erect, spindly and dwarfed**
- Leaves small, erect, **rolled and twisted**
- Tubers elongated with **tapering ends**
- Tuber eyes - numerous, shallow and more conspicuous

Potato spindle tuber viroid (PSTVd)

Survival and spread

- ***Primary***: Virus particles in infected tubers and mechanically through tuber cutting knives from infected to healthy tubers
- ***Secondary***: Virus particles transmitted by pollen and contaminated mouth parts of beetles, fleas and bugs

Potato spindle tuber viroid (PSTVd)

Management

- Use of PSTVd free potato seed tubers
- Sterilisation of cutting knives

Potato witches broom - Phytoplasma

Potato witches broom - Phytoplasma - strains

Symptoms

- Plants bushy - broom like, small, chlorotic leaves
- Auxillary and basal shoots develop
- **No tubers production** or
- **Small and minute aerial tubers with hairy sprouts**

Potato witches broom - Phytoplasma - strains

- **Grafting, dodder, leaf hoppers** - *Orosius albicinctus*, *Alebroides scutellatus*

Potato witches broom - Phytoplasma - strains

Management

- Use of disease free seed tubers
- Vector control